# Telemetry Endpoint Manual

2026-05-04

# Telemetry Endpoint

The HTTP POST endpoint that ingests opt-in anonymous scores from Preston-Check scans. Data feeds the annual State of Fintech Security report and the peer-comparison percentiles in the SaaS dashboard.

## URL

```
https://preston-check-telemetry.preston-check-edge.workers.dev/
```

Single production endpoint. Standalone Cloudflare Worker, on the anonymity-clean account subdomain `preston-check-edge.workers.dev`. This is the URL `lib/telemetry.sh` defaults to.

A Pages Function alternative at `app.preston-check.com/api/v1/telemetry` was attempted and removed — Cloudflare's Pages Function method dispatch fought us in ways the standalone Worker doesn't. The Worker is simpler, faster, and verified end-to-end.

## Privacy contract

The endpoint validates the documented schema and stores nothing else. Specifically:

- **Accepts** ten documented fields (see Schema below)
- **Hashes** the IP address out of the rate-limit key after the rate- limit decision is made
- **Strips** the IP address from the stored record entirely
- **Never** accepts source code, file paths, file names, or specific check IDs

The privacy implementation is auditable in 30 lines of TypeScript at `workers/telemetry/src/index.ts` (and the equivalent Pages Function at `web/customer/functions/api/v1/telemetry/index.ts`).

# Schema

POST `Content-Type: application/json` with this exact payload shape:

```
{
  "version":   "1.7.5",
  "tier":      "free",
  "lang":      "typescript",
  "repo_hash": "sha256-hex-string-64-chars",
  "pass":      268,
  "fail":      7,
  "warn":      14,
  "skip":      5,
  "total":     294,
  "timestamp": "2026-05-04T12:00:00Z"
}
```

Validation rules:

| FIELD | TYPE | CONSTRAINT |
|---|---|---|
| `version` | string | <= 32 chars |
| `tier` | string | one of `free` / `pro` / `enterprise` |
| `lang` | string | <= 32 chars |
| `repo_hash` | string | exactly 64 hex chars ( `/^[a-f0-9]{64}$/` ) or literal `unhashable` |
| `pass`, `fail`, `warn`, `skip`, `total` | number | 0 ≤ n ≤ 100000 |
| `timestamp` | string | ISO 8601 UTC matching `\d{4}-\d{2}-\d{2}T\d{2}:\d{2}:\d{2}Z` |

Anything else returns **400 invalid payload**.

# Responses

| HTTP CODE | MEANING |
|---|---|
| 200 | `{"ok":true}` — payload accepted and queued for storage |
| 400 | `invalid json` or `invalid payload` — validation failure |
| 405 | `method not allowed` — anything other than POST or OPTIONS |
| 429 | `rate limit exceeded` — IP exceeded 1000 requests/hour |

CORS: `Access-Control-Allow-Origin: *` (overridable via the `ALLOW_ORIGIN` env var on the Worker). OPTIONS preflight is handled.

# Storage

Each accepted POST writes:

**KV (aggregations)** — five counters incremented per event, each keyed by date / tier / language. Plus a score histogram bucketed in deciles per language. KV reads serve the score-percentile features in the customer dashboard with sub-50ms latency.

**D1 (raw records)** — one row inserted into the `scans` table with the full payload + computed score. Schema:

```
CREATE TABLE scans (
  id          INTEGER PRIMARY KEY AUTOINCREMENT,
  repo_hash   TEXT NOT NULL,
  version     TEXT NOT NULL,
  tier        TEXT NOT NULL,
  lang        TEXT NOT NULL,
  pass        INTEGER NOT NULL,
  fail        INTEGER NOT NULL,
  warn        INTEGER NOT NULL,
  skip        INTEGER NOT NULL,
  total       INTEGER NOT NULL,
  score       INTEGER NOT NULL,
  timestamp   TEXT NOT NULL,
  inserted_at INTEGER DEFAULT (strftime('%s','now'))
);
CREATE INDEX idx_scans_repo_hash ON scans(repo_hash);
```

Retention: 90 days on raw rows. Aggregates kept indefinitely.

# Self-hosting

Air-gapped or on-premise customers can stand up their own endpoint and point the runner at it:

```
export PRESTON_TELEMETRY_ENDPOINT="https://telemetry.acme.internal/v1/ingest"
preston-check --telemetry-opt-in
```

The reference implementation (Worker + schema + workflow) lives in `workers/telemetry/` and is Apache 2.0. Roughly 30 minutes to fork + deploy to your own Cloudflare account.

# How customers opt in

Three equivalent paths, off by default:

```
preston-check --telemetry-opt-in            # per-run flag
export PRESTON_TELEMETRY=1                  # env var
echo 'telemetry: opt_in' >> .preston-check.yml   # config file
```

`--airgap` overrides every opt-in.

# Querying the dataset

Operator-only — requires Cloudflare API token:

```
export CLOUDFLARE_API_TOKEN="..."
export CLOUDFLARE_ACCOUNT_ID="..."

# Total scan count
wrangler d1 execute preston-check-telemetry --remote --command \
  "SELECT count(*) FROM scans"

# Daily volume last 7 days
wrangler d1 execute preston-check-telemetry --remote --command \
  "SELECT substr(timestamp,1,10) AS day, count(*) FROM scans
   WHERE timestamp > datetime('now','-7 days') GROUP BY day"

# Score distribution by language
```

```
wrangler d1 execute preston-check-telemetry --remote --command \
  "SELECT lang, avg(score) FROM scans GROUP BY lang ORDER BY avg(score) DESC"
```

# Source

```
workers/telemetry/                              standalone Worker (production
endpoint)
  src/index.ts                                  TypeScript Worker implementation
  schema.sql                                    D1 schema (already applied)
  wrangler.toml                                 project config + bindings
  README.md                                     setup notes
.github/workflows/telemetry-deploy.yml          CI deploy on push

web/customer/functions/api/v1/telemetry/        Pages Function (alternative
endpoint)
  index.ts
```

# Cross-links

- **Privacy + telemetry overview**: `docs/telemetry.md`
- **Worker setup**: `workers/telemetry/README.md`
- **Customer Portal manual**: `docs/manuals/customer-portal.md`